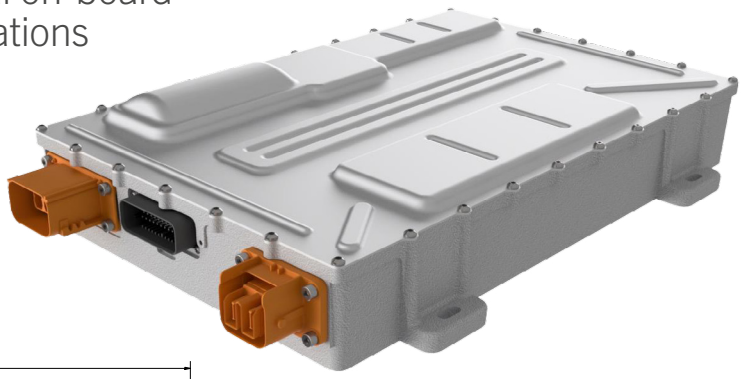
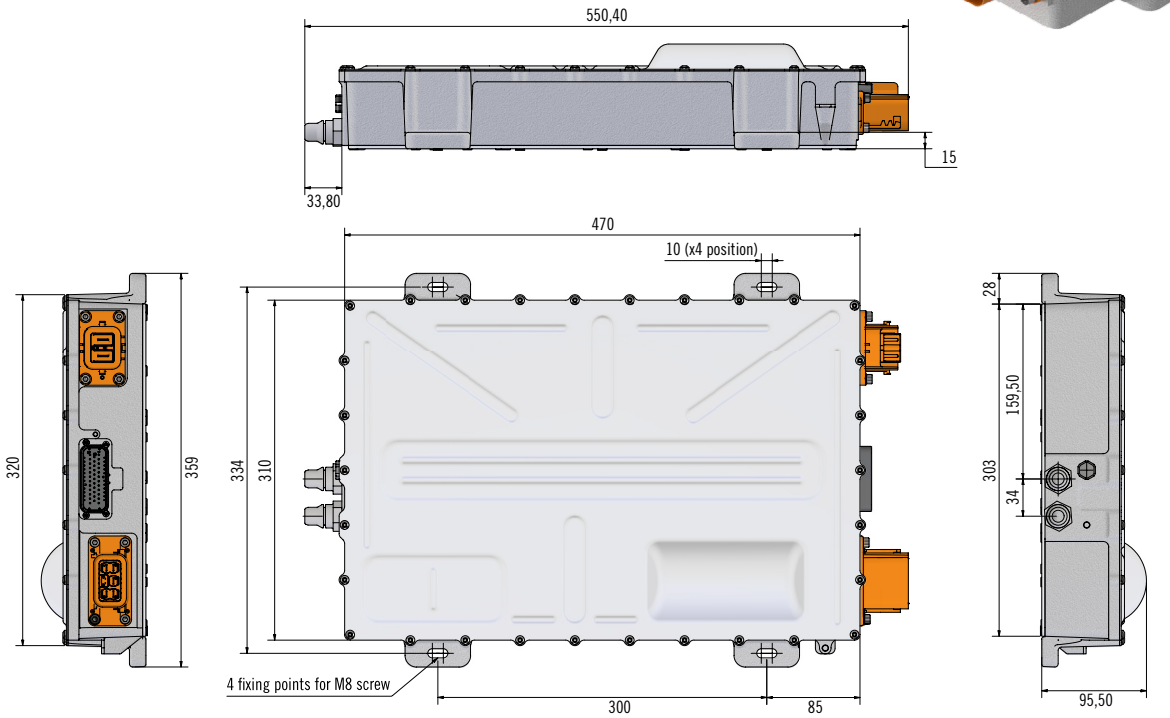
AN MTA BRAND

BHP22KL

22 kW single and three phase bidirectional on-board charger for heavy duty and rugged applications





APPLICATIONS

BHP22KL charger series is a 22 kW bidirectional high power density power module, very versatile, safe, optimized for on-board battery powered vehicle applications. The module is based on the latest SIC semiconductors and power converter technologies which results in high efficiency and prime performance in all harsh applications.
The BHP22KL can be plugged as single or three phase AC and works in “Forward Charger Mode” or “Reverse Power Mode” (bidirectional operations) to perform V2L applications.
The BHP22KL has been successfully tested to meet international EMC, safety and environmental standards.

FEATURES

- 7.3 kW AC single and 22 kW three-phase bidirectional operation
- Parallelable in charge mode and in V2L operation
- AC charger – SAE J1772 and EN 61851 compliance
- DC fast charging (CCS) communication – DIN 70121, ISO 15118
- AC plug locking/unlocking management
- AC and HVDC interlock monitoring
- Typical efficiency >94%
- AC/DC galvanic isolation
- UDS, functional safety (ASIL-B), cybersecurity compliant
- IP67, IP6K9K protection rating
- HVDC outputs are short-circuit protected
- Fully CAN monitored and controlled (v2.0B)
- SAE J1939 compliant
- UL1741, IEEE1547 compliant



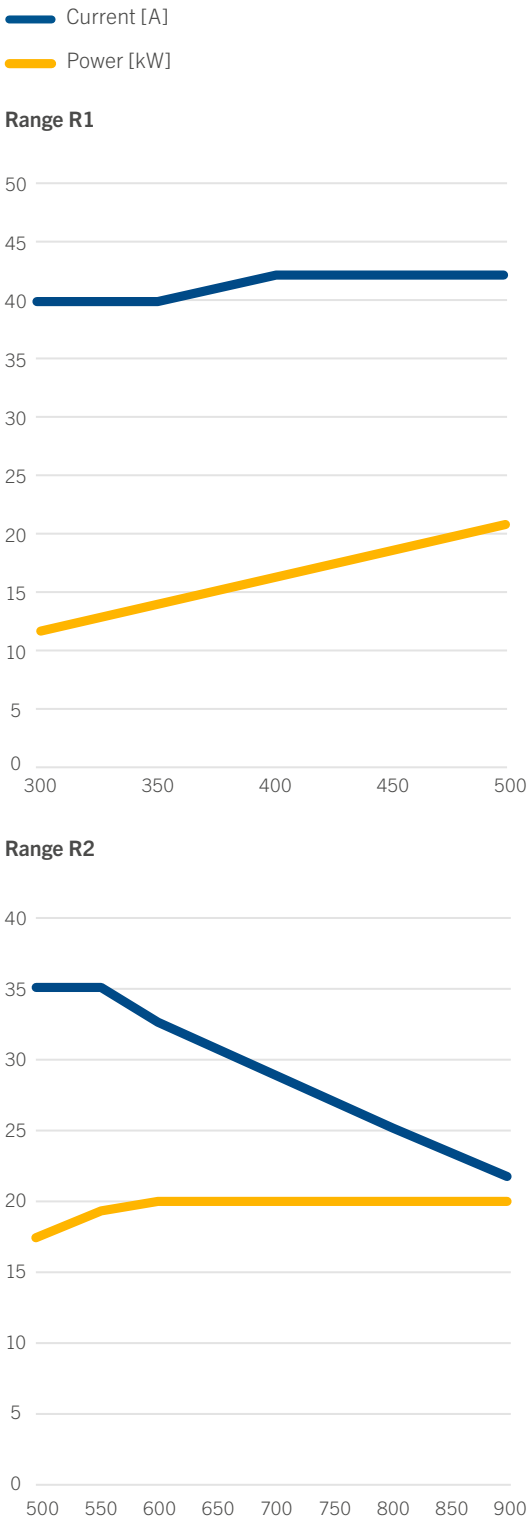
ACCESSORIES

HBI22D-05	AC and LV mating cable 5 m
HBO22D-05	HVDC mating cable 5 m

SPECIFICATIONS

AC input data (charge mode)		Units
1-phase AC voltage (phase - neutral)	195 ÷ 265	Vac
3-phase AC voltage (phase - phase)	365 ÷ 440	Vac
Line frequency	47 ÷ 63	Hz
AC input current (max)	32	Aac
Absorbed apparent power (max)	22 (7.3 kVA in single phase)	kVA
Power factor	>0.98	-
AC current THD	<10	%
AC output Reverse Power Mode (V2L)		
1-phase AC voltage	230/240 (split voltage)	Vac
3-phase AC voltage (phase-phase)	400/420	Vac
Line frequency	50/60	Hz
AC voltage THD	<3	%
AC output current (max)	32	Aac
Apparent power (max)	9	kVA
Battery voltage range	350 ÷ 500 (range R1) 550 ÷ 900 (range R2)	Vdc
HVDC output Forward Charge Mode		
V01 HV output voltage	0 ÷ 500 (range R1) 0 ÷ 900 (range R2)	Vdc
V01 HV voltage accuracy (typ)	±0.5	%
V01 HV output current (max)	42 (range R1) 35 (range R2)	Adc
V01 HV rated output power (max)	>20 (7.0 kW in single phase)	kW
General data		
Ambient temperature (operating)	-40 ÷ +85	degC
Ambient temperature (full performance)	-40 ÷ +70	degC
Power de-rating	+50 (of the coolant liquid)	degC
Heat dissipation	Liquid cooling (50:50% Glycol/Water, F > 6 lt/min)	-
Coolant pressure loss at 6 lt/min, 25 °C	<200	mbar
Protection degree	IP67/IP6K9K	-
Efficiency	>94 at from 50% to max load	%
Communication and Control interface	CAN V2.0B - SAE 1939 - UDS	
Standard applied		
Environment	ISO 16750-4, SAE J1455	-
EMC – Emission and immunity	ECE regulation 10	-
Safety	ECE Regulation 100; SAE J2344; ISO 16750-2; ISO 26262 Functional Safety (ASIL-B); ISO SAE21434 (Secure Boot, Secure Programming)	-
Insulation resistance	HVDC output/PE: >10 MΩ at 1000 Vdc	-
Touch current	<3.5	mA
Chassis ground stud	Directly on the chassis	-
Mechanical data		
Dimensions (width x depth x height)	470x310x95	mm
Weight	17.7	kg
Case material	Aluminum alloy	-

CHARGE MODE

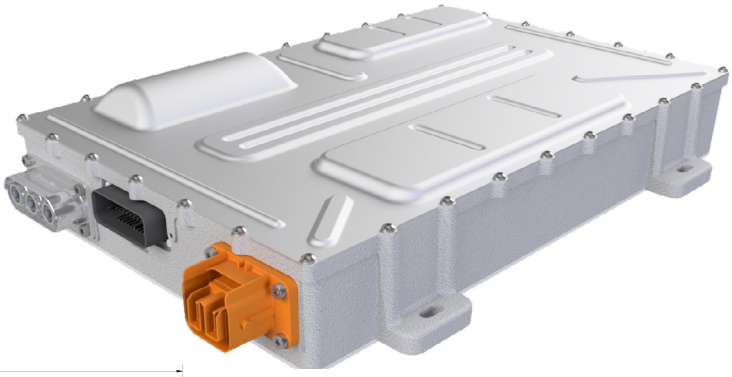


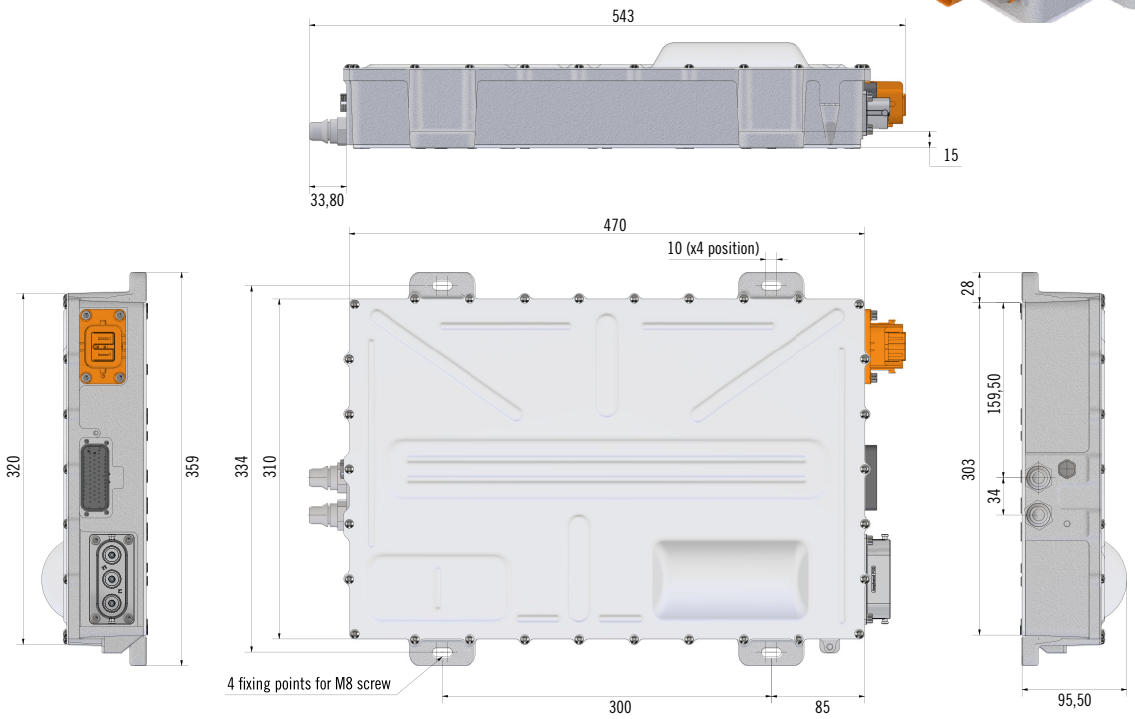
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BHP19KL

19.2 kW single phase bidirectional on-board charger
for heavy duty and rugged applications





APPLICATIONS

BHP19KL charger series is a 19.2 kW bidirectional high power density power module, very versatile, safe, optimized for on-board battery powered vehicle applications. The module is based on the latest SIC semiconductors and power converter technologies which results in high efficiency and prime performance in all harsh applications.
The BHP19KL can be configured in “Forward Charger Mode” or “Reverse Power Mode” (bidirectional operations) to perform V2L applications.
The BHP19KL has been successfully tested to meet international EMC, safety and environmental standards.

FEATURES

- 19.2 kW AC single bidirectional operation
- Parallelable in charge mode and in V2L operation
- AC charger – SAE J1772 and EN 61851 compliance
- DC fast charging (CCS) communication) – DIN 70121, ISO 15118
- AC plug locking/unlocking management
- AC and HVDC interlock monitoring
- Typical efficiency >94%
- AC/DC galvanic isolation
- UDS, functional safety (ASIL-B), cybersecurity compliant
- IP67, IP6K9K protection
- HVDC outputs are short-circuit protected
- Fully CAN monitored and controlled (v2.0B)
- SAE J1939 compliant
- UL1741, IEEE1547 compliant



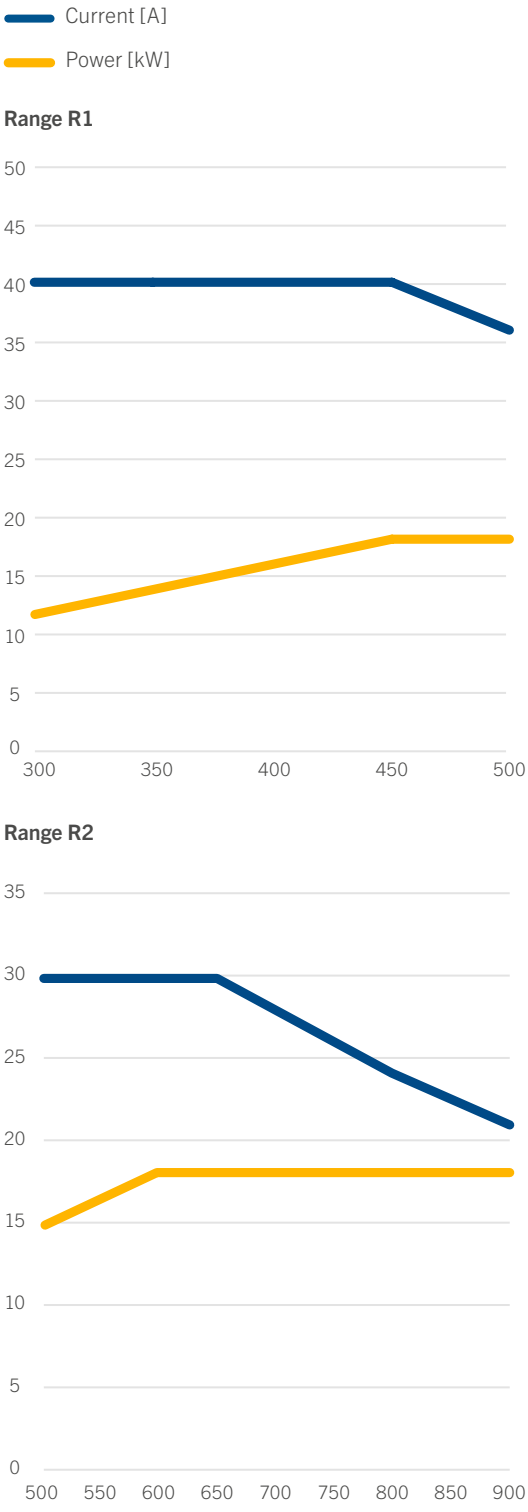
ACCESSORIES

HBI19D-05	AC and LV mating cable 5 m
HBO19D-05	HVDC mating cable 5 m

SPECIFICATIONS

AC input data (Forward Charge Mode)		Units
1-phase AC voltage	90 ÷ 265	Vac
Line frequency	47 ÷ 63	Hz
AC input current (max)	80	Aac
Absorbed apparent power (max)	19.2	kVA
Power factor	>0.98	-
AC current THD	<10	%
AC output reverse power mode (V2L)		
1-phase AC voltage	120/230/240 (split voltage)	Vac
Line frequency	50/60	Hz
AC voltage THD	<3	%
AC output current (max)	80	Aac
Apparent power (max)	9	kVA
Battery voltage range	350 ÷ 500 (range R1) 550 ÷ 900 (range R2)	Vdc
HVDC output Forward Charge Mode		
V01 HV output voltage	0 ÷ 500 (range R1) 0 ÷ 900 (range R2)	Vdc
V01 HV voltage accuracy (typ)	±0.5	%
V01 HV output current (max)	40 (range R1) 30 (range R2)	Adc
V01 HV rated output power (max)	>18	kW
General data		
Ambient temperature (operating)	-40 ÷ +85	degC
Ambient temperature (full performance)	-40 ÷ +70	degC
Power de-rating	+50 (of the coolant liquid)	degC
Heat dissipation	Liquid cooling (50:50% Glycol/Water, F > 6 lt/min)	-
Coolant pressure loss at 6 lt/min, 25 °C	<200	mbar
Protection degree	IP67/IP6K9K	-
Efficiency	>94 at from 50% to max load	%
Communication and Control interface	CAN V2.0B - SAE 1939 - UDS	-
Standard applied		
Environment	ISO 16750-4, SAE J1455	-
EMC – Emission and immunity	ECE regulation 10	-
Safety	ECE Regulation 100; SAE J2344; ISO 16750-2; ISO 26262 Functional Safety (ASIL-B); ISO SAE21434 (Secure Boot, Secure Programming)	-
Insulation resistance	HVDC output/PE: >10 MΩ at 1000 Vdc	-
Touch current	<3.5	mA
Chassis ground stud	Directly on the chassis	-
Mechanical data		
Dimensions (width x depth x height)	470x310x95	mm
Weight	17,7	kg
Case material	Aluminum alloy	-

CHARGE MODE



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